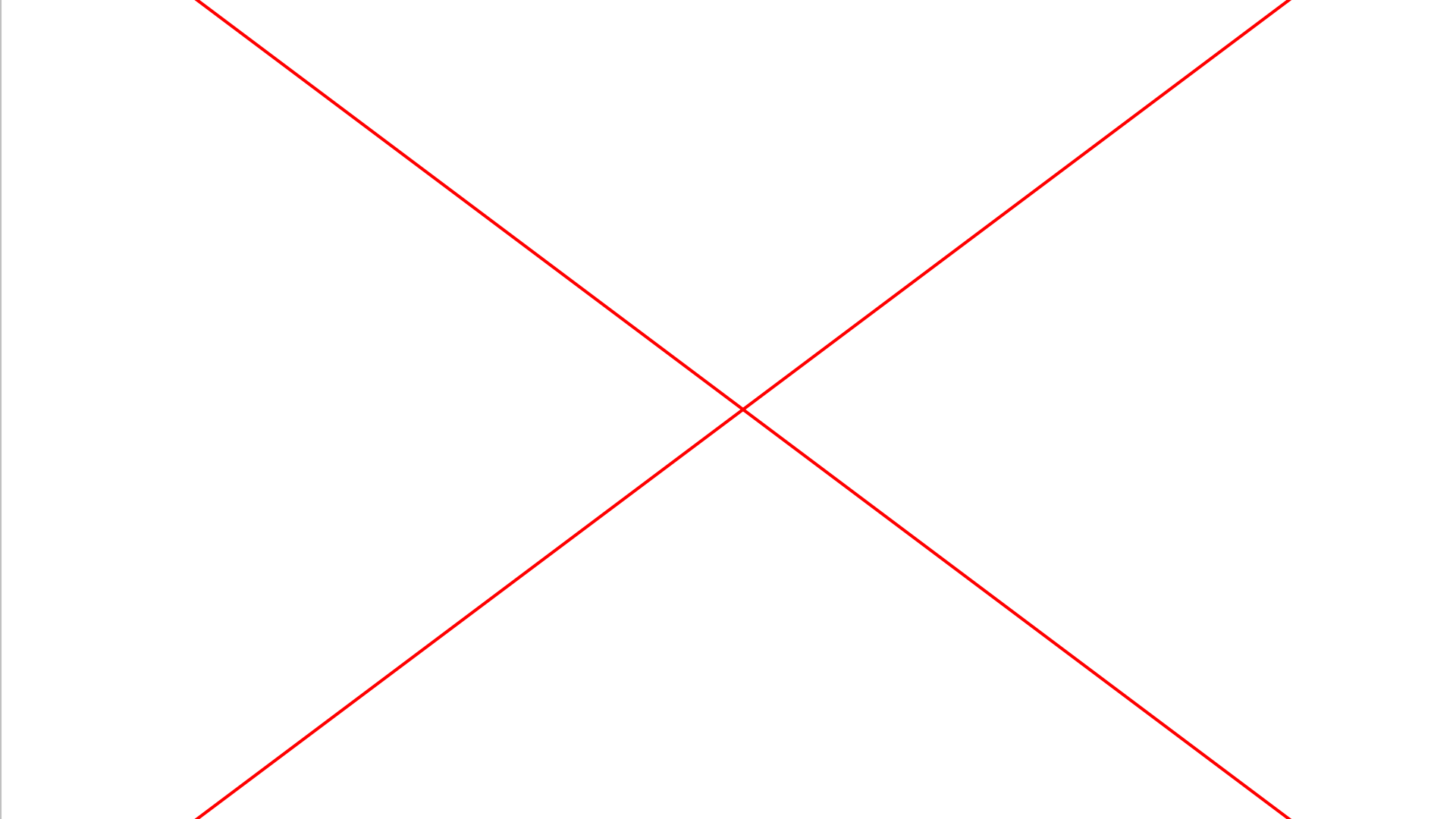








**How might we design a Spider Web-Inspired Tensegrity Robotic System that acts as a protective shield for aerial aid delivery ?**



# AIDtensegrity - Background Research

Comparison of Aerial Delivery Systems



Deployment Versatility

Diverse Environment



System Performance Spectrum

Low-tech / Basic

High-tech / Reusable



Specific Environment





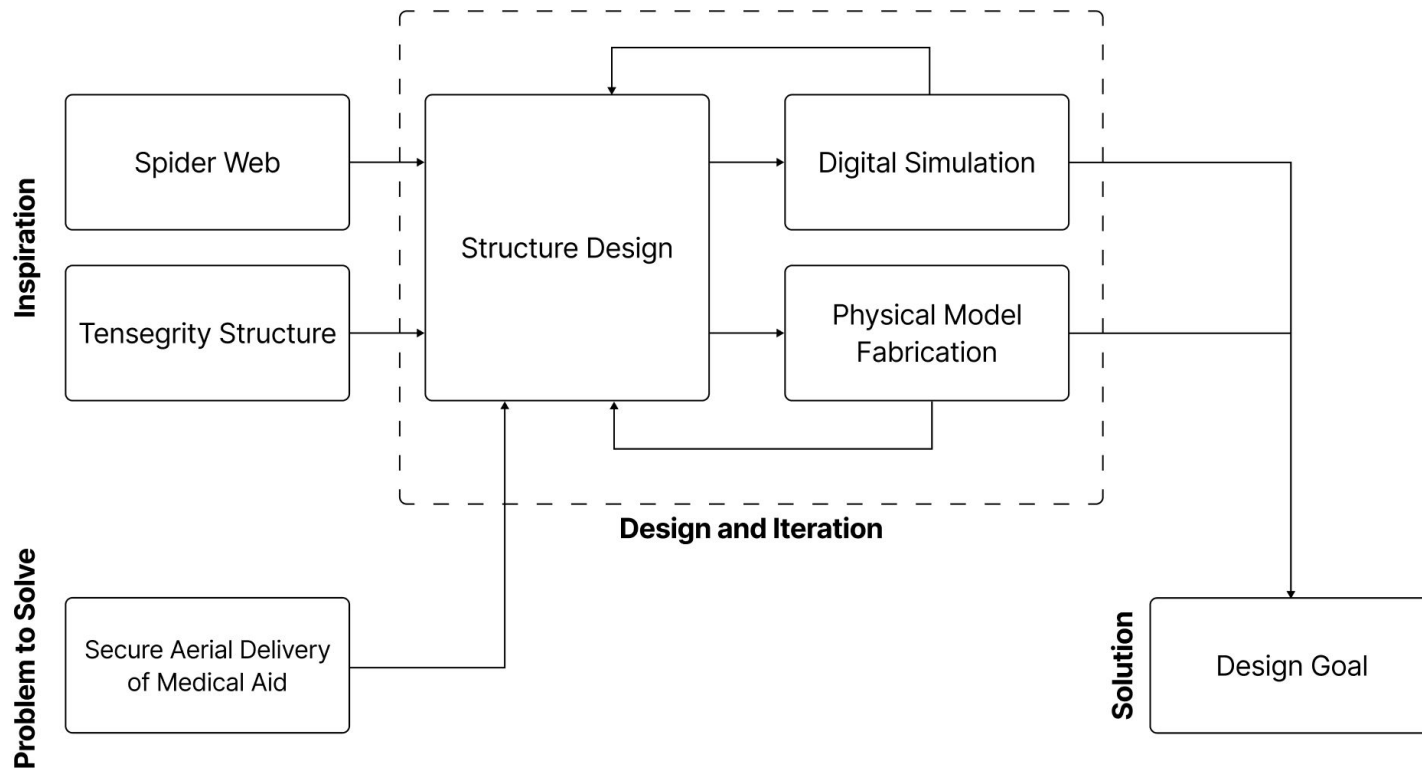
## AIDtensegrity - Design Goal and Strategy



Our objective is to develop an aerial delivery system capable of transporting fragile medical aid to disaster-stricken areas. The system is designed to support a **payload of 2–4 kg**, fit within a **50–100 cm size** range, and withstand impact from a **30-meter drop**, ensuring the safe and intact delivery of essential supplies under challenging conditions.



## AIDtensegrity - System Diagram





## AIDtensegrity - Method Diagram

### Design Parameters

#### Structure Design

**Node design (Protection):** Safeguarding the system to prevent collapse and ensure it can survive and protect the internal payload from external impacts.

**Structure design (Lightweight):** Using minimal material while maintaining structural integrity

**Payload design (Payload accessibility):** Optimized internal space for easy payload access and use.

#### Physical Model Fabrication

**Fabrication Process (Construction time):**  
The time required to assemble the structure.

**Material Decision (Cost of production):**  
Total expense involved in manufacturing and assembling the structure.

### Methodology

#### Digital Simulation

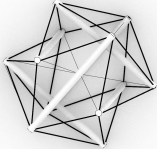
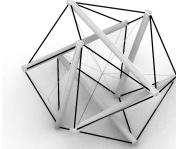
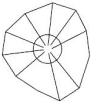
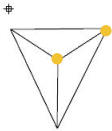
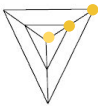
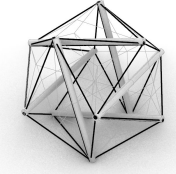
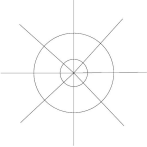
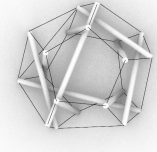
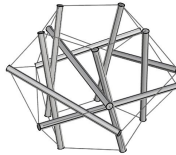
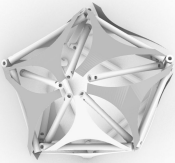
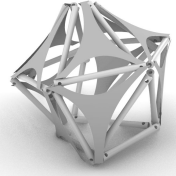
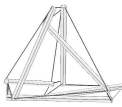
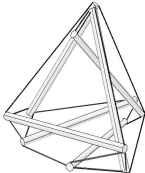
Calculate the tensions acting on each node.

Determine the relationship between weight and size.

Figure out what kind of material we can use for fabrication.



# Design

| DESIGN AIM/TYPE                                      | STRUCTURE DETAIL                              | DESIGN PROCESS                                                                     |                                                                                    |                                                                                     |                                                                                      | TOP VIEW                                                                             | PERSPECTIVE VIEW                                                                     |
|------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Squishy robot structure design                       | Cables segments-32<br>Nodes- 61<br>Struts- 6  |    |    |   |    |    |    |
| Spider web node configuration applied to tensegrity. | Cables segments-49<br>Nodes- 74<br>Struts- 6  |   |   |  |   |   |   |
| Morphed web applied as tension system                | Cables segments-42<br>Nodes- 62<br>Struts- 10 |   |   |   |   |   |   |
| Optimized 30-strut tensegrity                        | Cables segments-60<br>Nodes- 120<br>Strut- 21 |   |   |   |   |   |   |
| Optimized 6-strut tensegrity                         | Cables segments-18<br>Nodes- 60<br>Struts- 6  |  |  |  |  |  |  |



(Squishy 1kg, 1 meter)



(Web-based 1kg, 1 meter)



(10 Struts 1kg,  
1 meter)

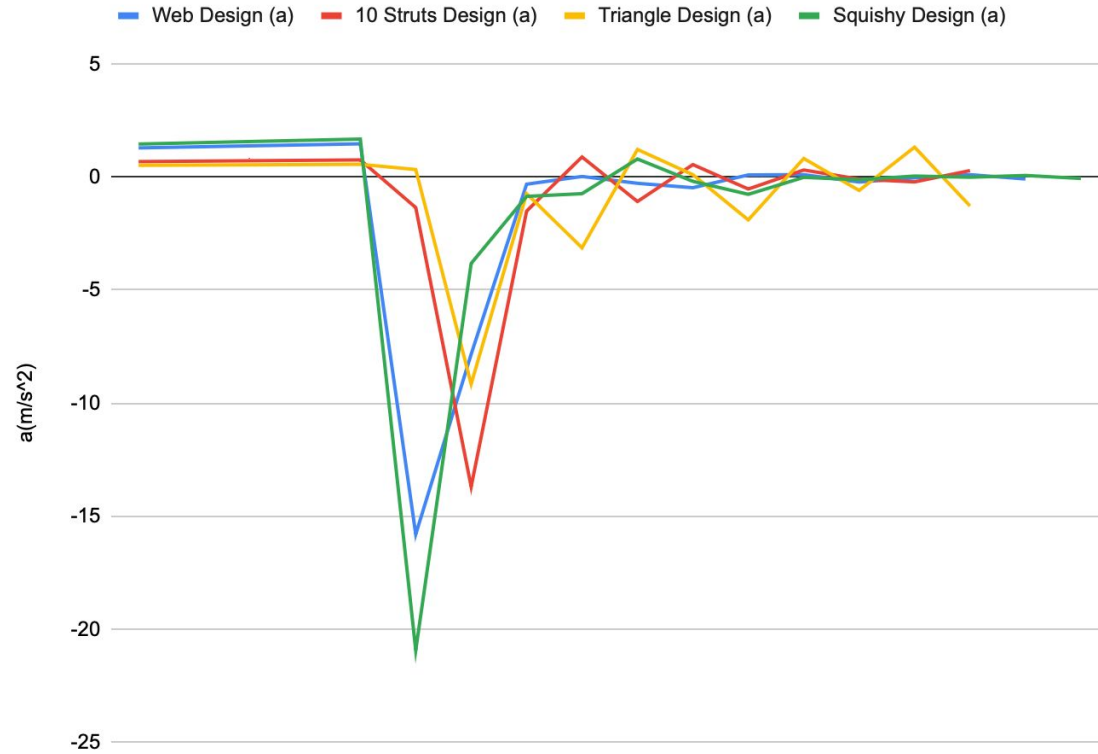


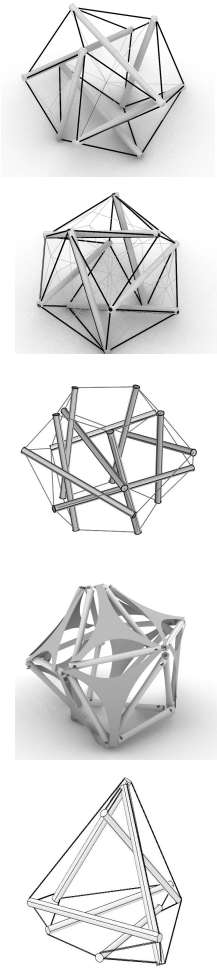
(Triangle 1kg,  
1 meter)



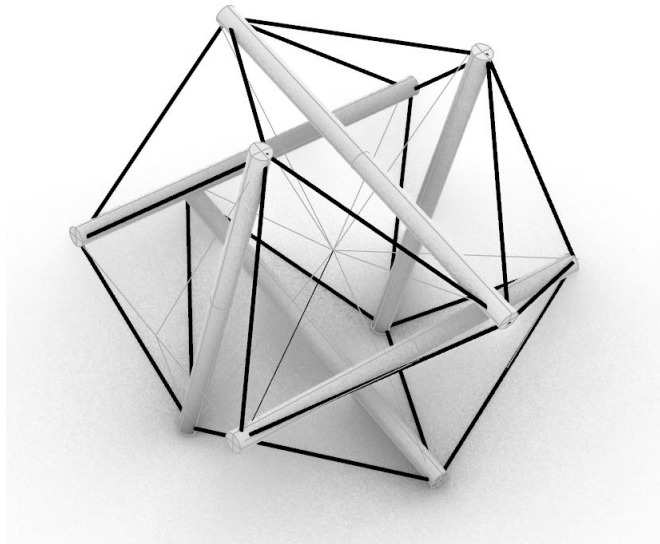
# AIDtensegrity - Design & Methodology

Comparison of Different Systems (Best Case)



|                                                                                   |   | Lightweight                                | Construction Time | Cost of Production                | Payload accessibility & Inner space availability | Stability/Protection                        |
|-----------------------------------------------------------------------------------|---|--------------------------------------------|-------------------|-----------------------------------|--------------------------------------------------|---------------------------------------------|
|  | A | 2 (Heavier than E)                         | 1 (Fastest)       | 4                                 | 4                                                | 3                                           |
|                                                                                   | B | 3 (Heavier than C)                         | 2                 | 3                                 | 4                                                | 2                                           |
|                                                                                   | C | 4 (Heavier than E)                         | 5                 | 2                                 | 2                                                | 5                                           |
|                                                                                   | D | 5 (The heaviest of all the five structure) | 4                 | 1 ( Most expensive to construct ) | 1 ( provides the largest usable interior space ) | 1 ( D is stable and offers more protection) |
|                                                                                   | E | 1 (The lightest of all the five structure) | 3                 | 5                                 | 3                                                | 4                                           |

Based on the parameter results, we selected this tensegrity structure for production.



6 strut Tensegrity Structure (Icosahedron)



# Testing & Analysis



# AIDtensegrity - Design & Methodology

Comparison of Aerial Delivery Systems (Squishy)

Comparison Graph of Payload Acceleration (Force) for Two Designs During a 30m Drop



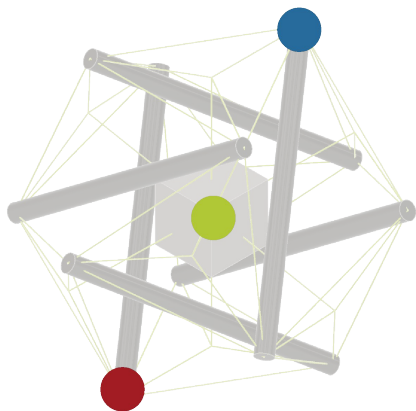
The **web-based design absorbs more impact than** squishy-based design.



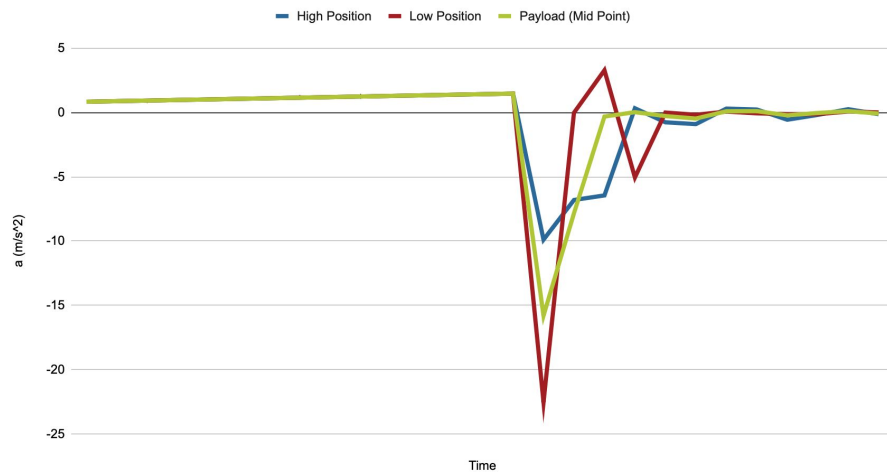
(Squishy 1kg, 1 meter) (Web-based 1kg, 1 meter)



## AIDtensegrity - Design & Methodology



Comparison Graph of Payload Acceleration (Force) at Three Different Points in Web-Based Designs During a 30m Drop

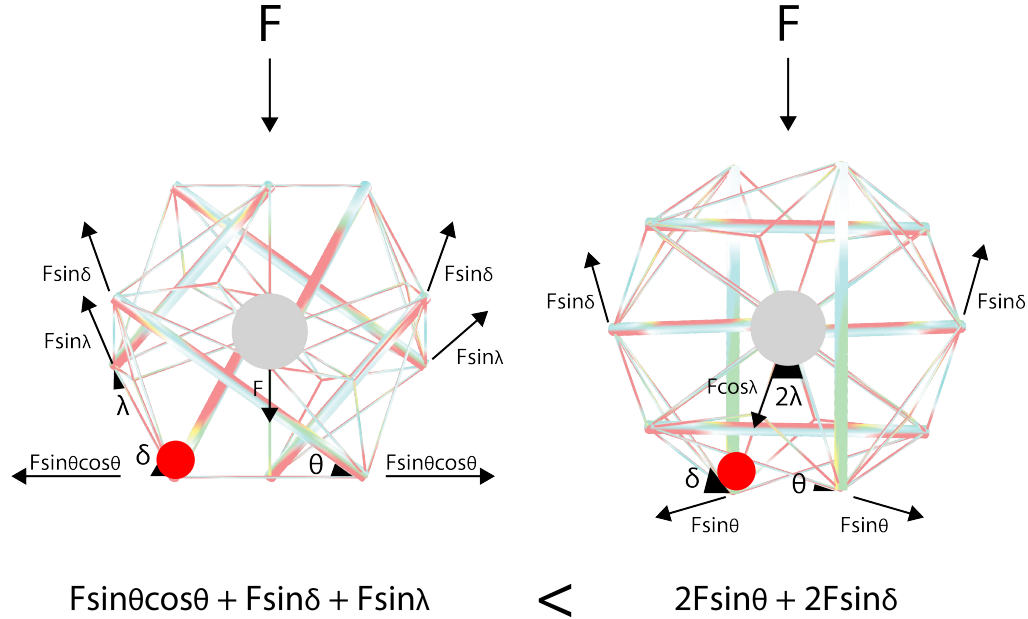


The payload experiences less impact than the lowest point, which directly contacts the ground. This design **effectively protects the payload and reduces impact.**



# AIDtensegrity - Design & Methodology

Tensions act on the nodesS



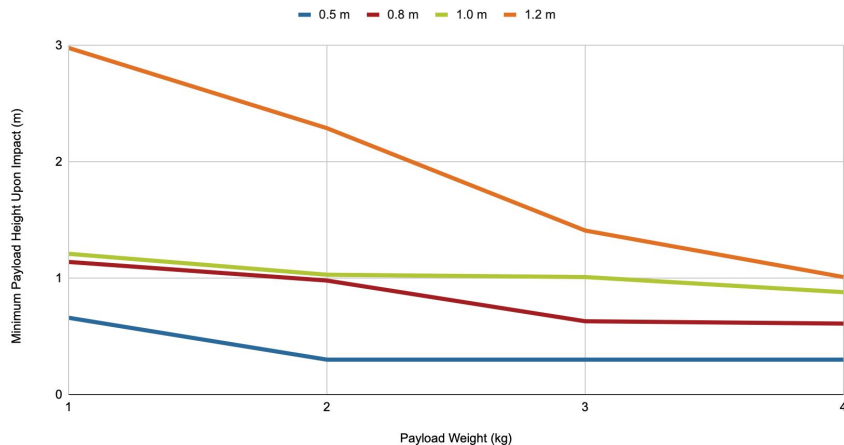
All angles should be greater than  $0^\circ$  and less than  $90^\circ$ . In this case, the **Best Case** always experiences less force than the **Worst Case**.



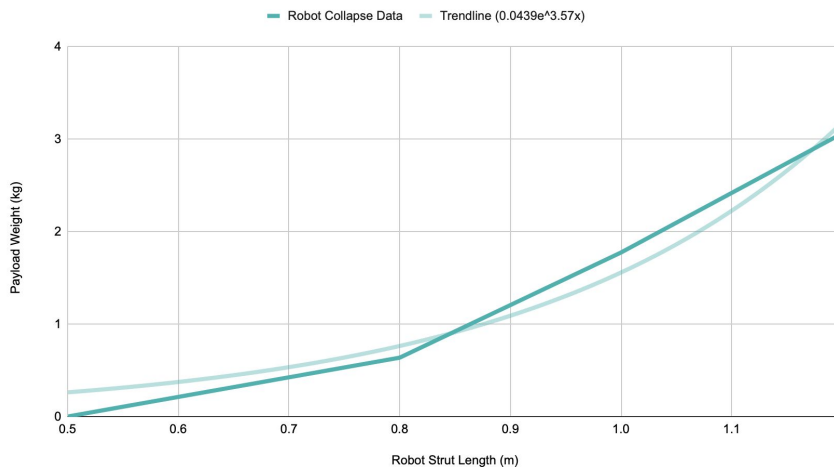


## AIDtensegrity - Design & Methodology

Web-Based Simulation Data for Payload Weight vs. Minimum Payload Height at Various Strut Lengths During a 30m Drop



Graph of Payload Weight vs. Strut Length During a 30m Drop



By adjusting the payload weight and robot size, we can derive the relationship equation between these two features.

$$0.0439 e^{3.57x}$$



## AIDtensegrity - Design & Methodology - Conclusion

1

Web-based design **effectively protects the payload and reduces impact.**

2

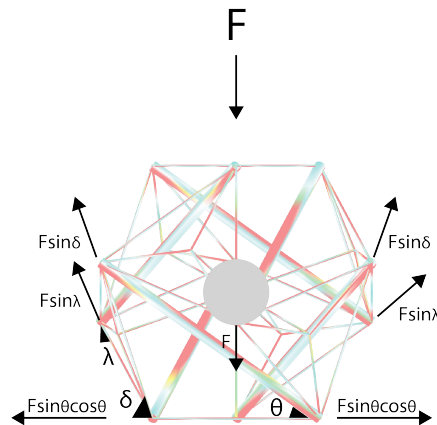
$$0.0439 e^{3.57x}$$

Based on the relationship equation, for this design, if we want to carry a **2–4 kg payload**, the **strut length should be between 1.07 and 1.26 meters.**

3

$$(F \sin \theta) \cos \theta + F \sin \delta + F \sin \lambda$$

When the **angle is smaller**—meaning more struts are added to create a smoother, ball-like shape and increase the internal space—the **strings experience less tension.**

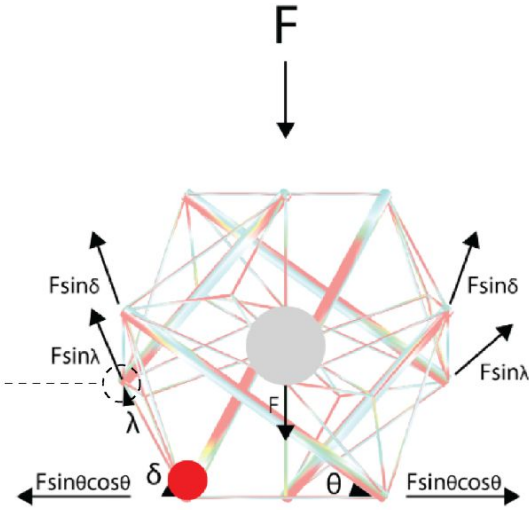
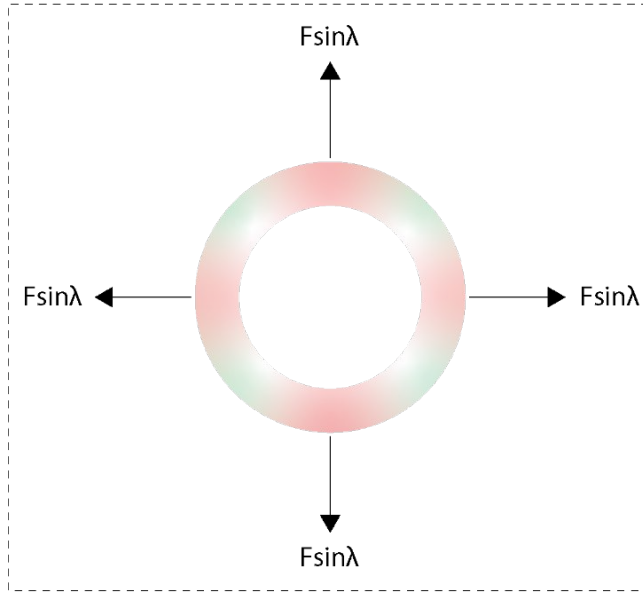




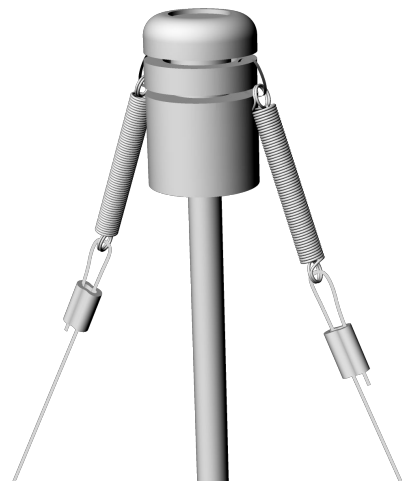
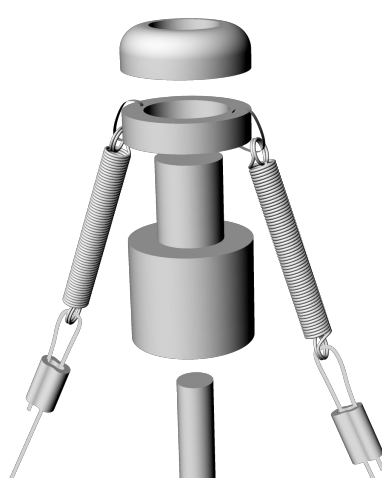
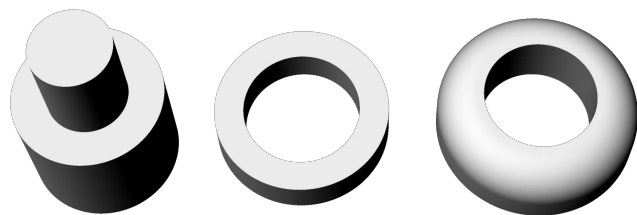
# Node Design



## AIDtensegrity - Node Design (Physics Analysis)



$$F\sin\theta\cos\theta + F\sin\delta + F\sin\lambda$$





## AIDtensegrity - Node Design (Material)



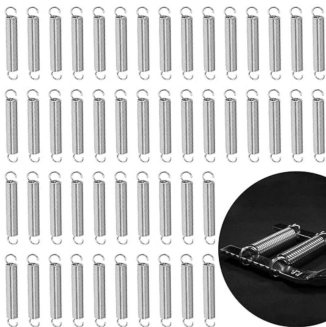
**Wire Rope**

Working load limit of 96 lb



**Crimping Sleeve**

Aluminum construction



**Spring**

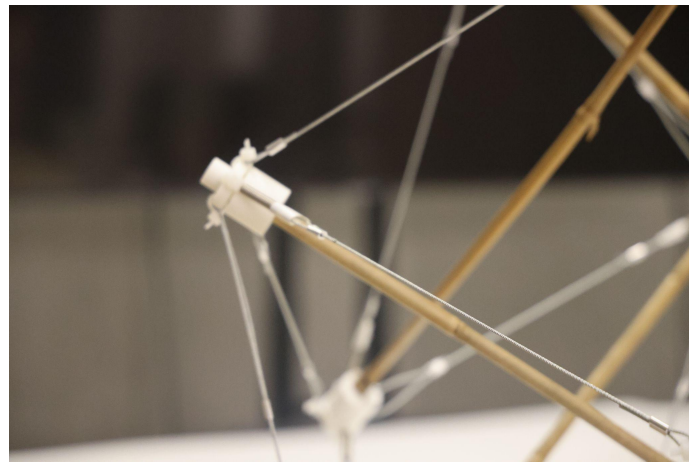
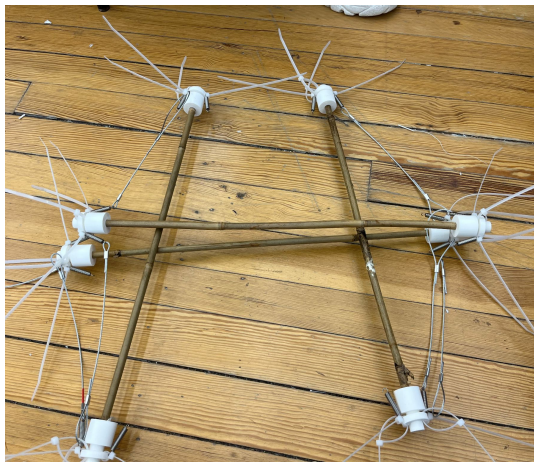
Maximum Safe Load is 8.46 lb



**Plastic Strip**

Tensile strength: 90 lb

## AIDtensegrity - Node Design (Fabrication)





# Payload Design



## AIDtensegrity - Payload Design (Inspiration)

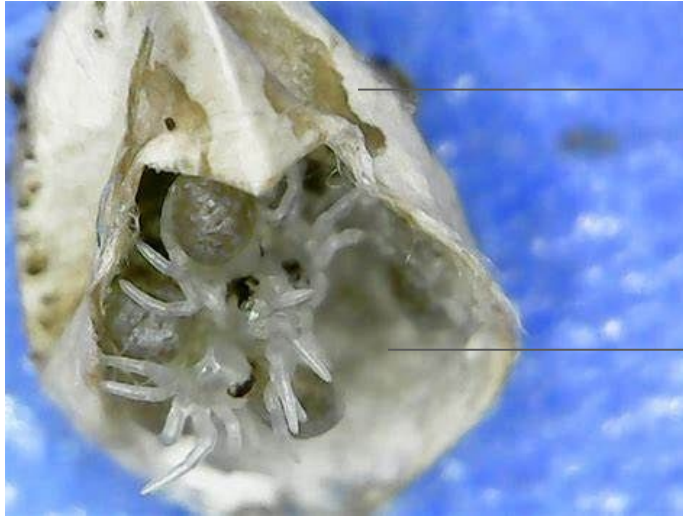


Spoke

Spider egg sac



## AIDtensegrity - Payload Design (Inspiration)

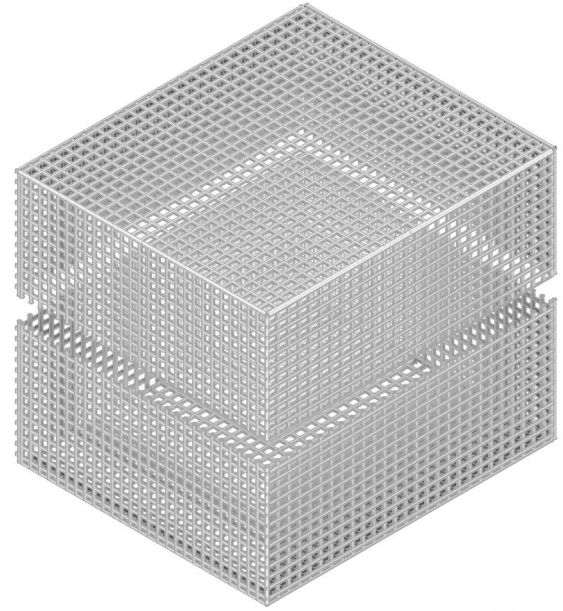


Ampullate Silk ( tough and elastic)

Tubiliform Silk (foam like)



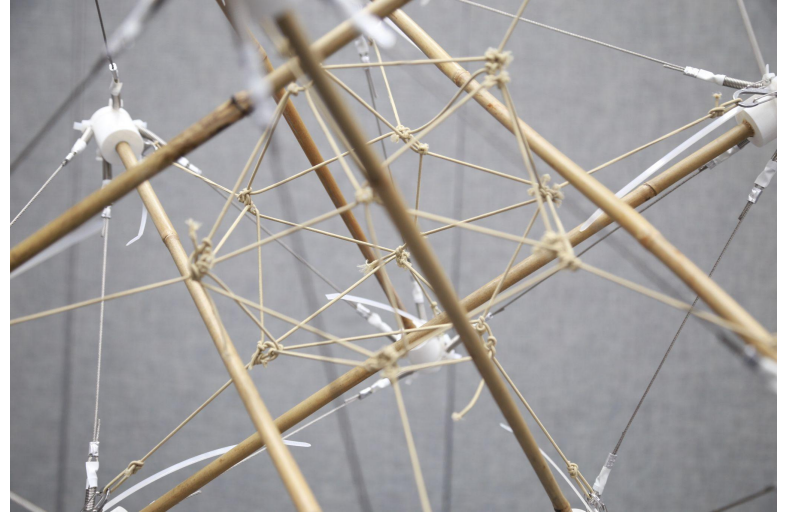
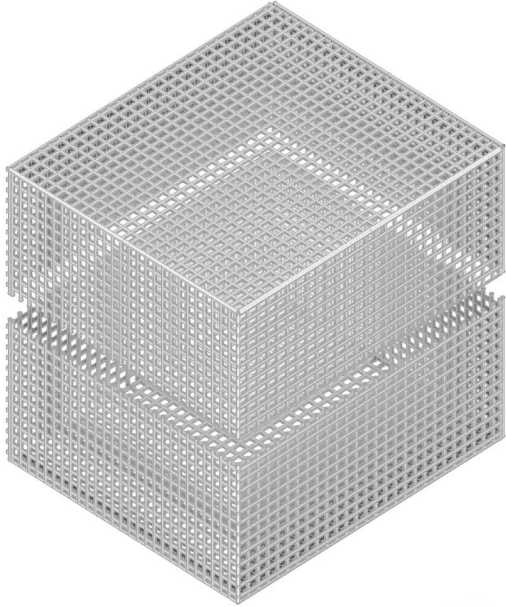
## AIDtensegrity - Payload Design (implementation)



Wire mesh



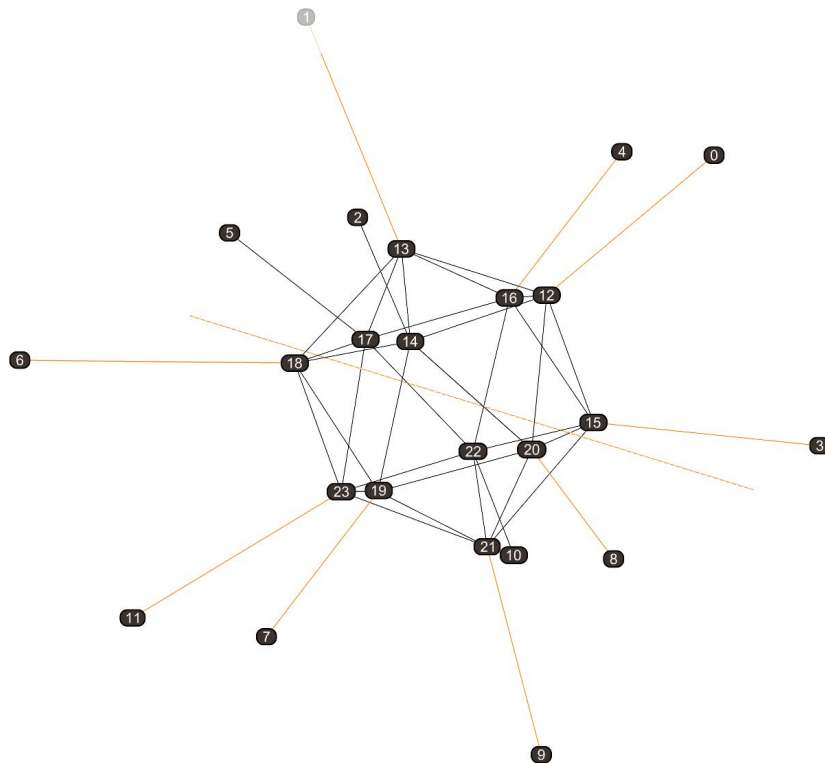
## AIDtensegrity - Payload Design (implementation)



Replacing mass with materiality and rigidity with flexibility- a design logic inspire by spider egg sac



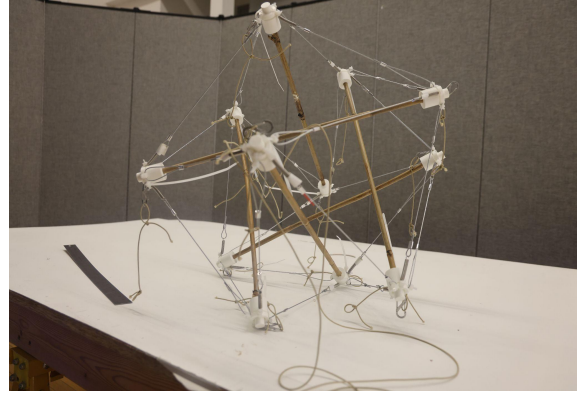
## AIDtensegrity - Payload Design (implementation)



Payload fabrication- Inspired by spider construction process.

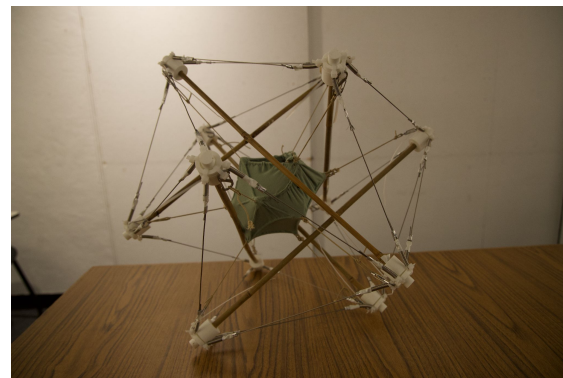


## AIDtensegrity - Payload Design (Fabrication)





## AIDtensegrity - Payload Design (Fabrication)



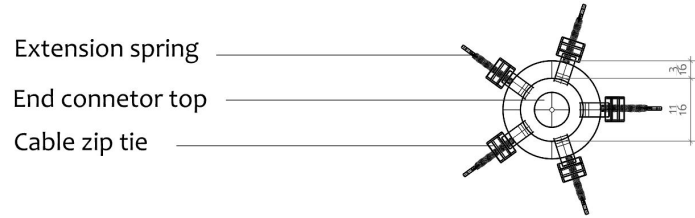


# Fabrication Iteration

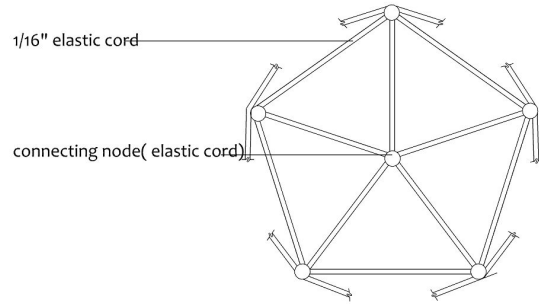




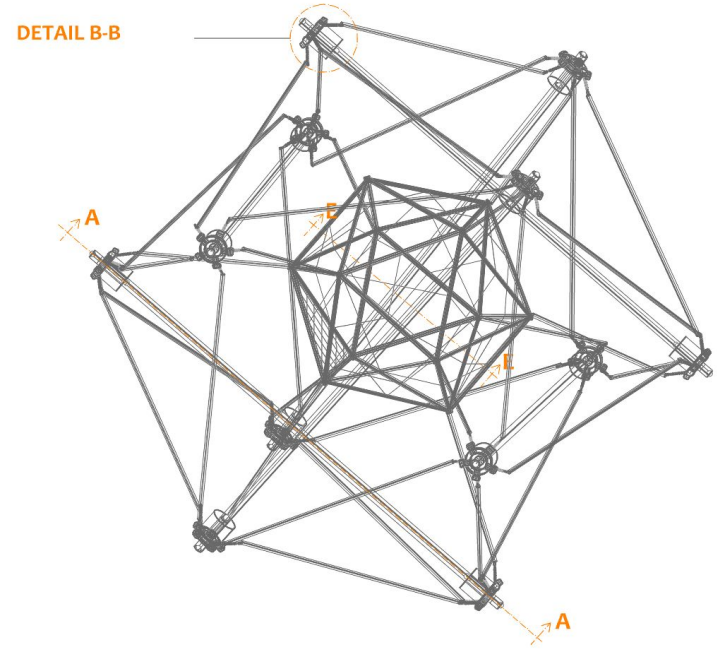
## AIDtensegrity - Fabrication (Section Drawing)



**DETAIL B-B**



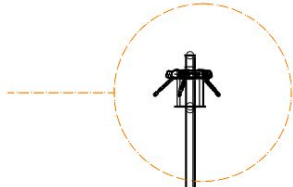
**DETAIL E-E**



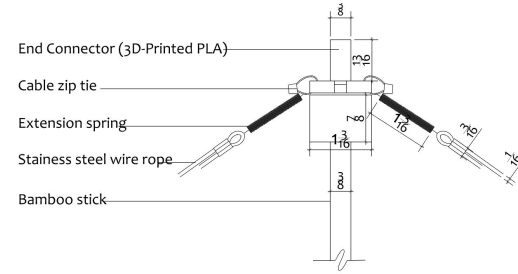
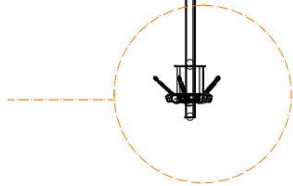


## AIDtensegrity - Fabrication (Section Drawing)

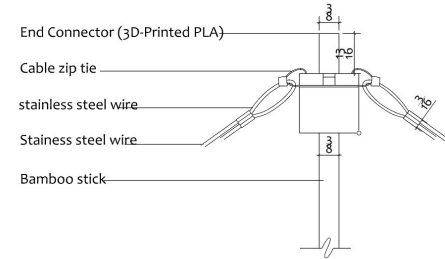
DETAIL C-C



DETAIL D-D



DETAIL C-C

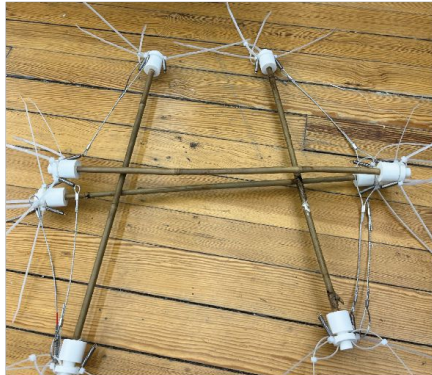


DETAIL D-D

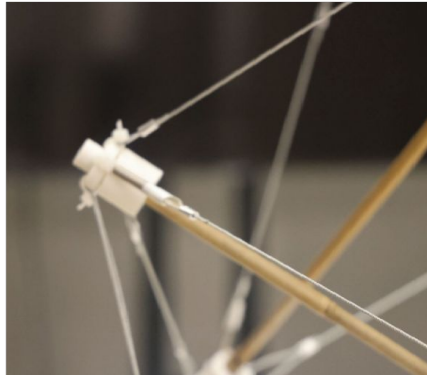
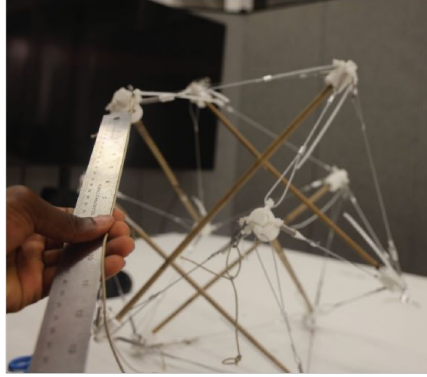


# AIDtensegrity - Fabrication

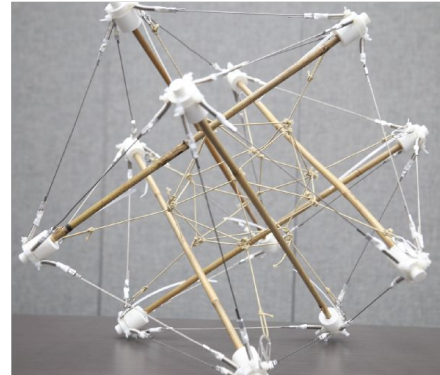
Step 1: Model Fabrication



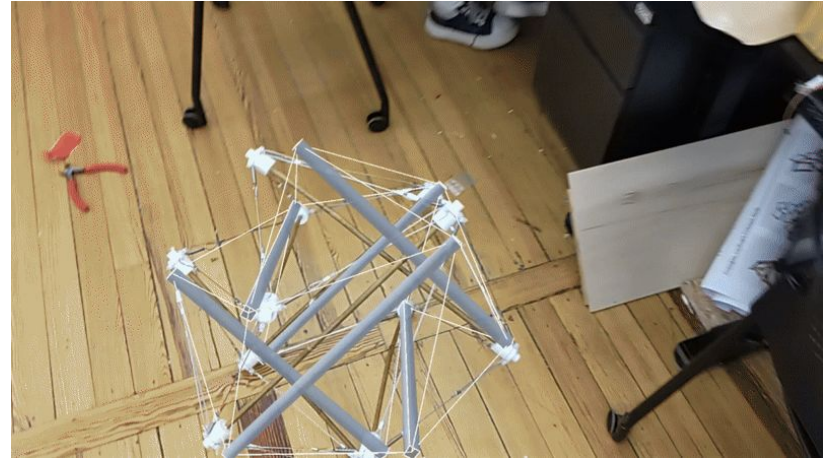
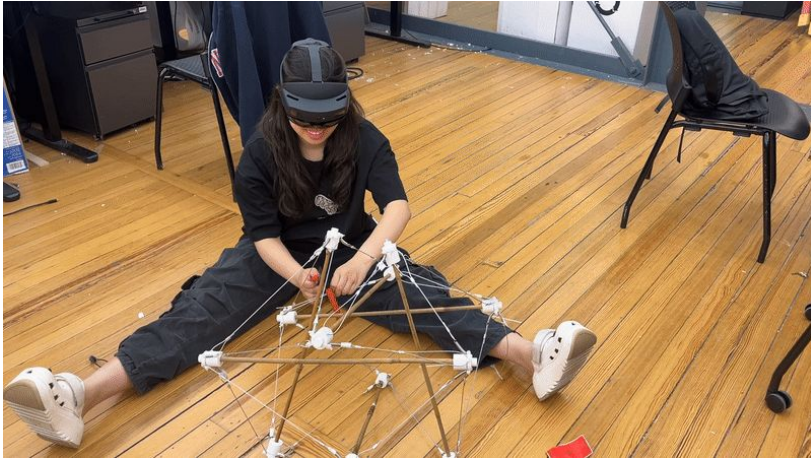
Step 2: Node Fabrication



Step 3: Payload Fabrication



## Step 4: Using Hololens to Refine Model



# AIDtensegrity - Fabrication

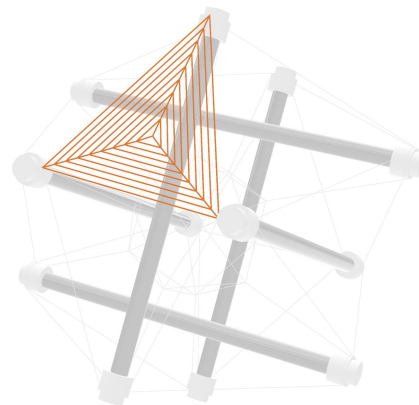
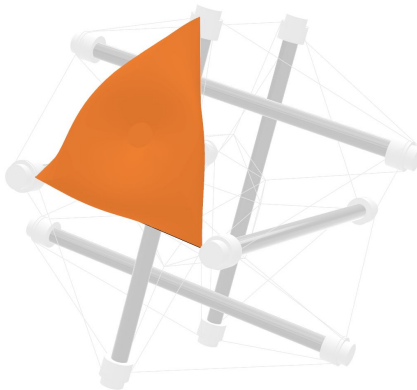
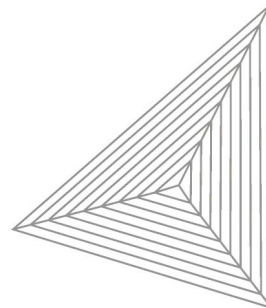
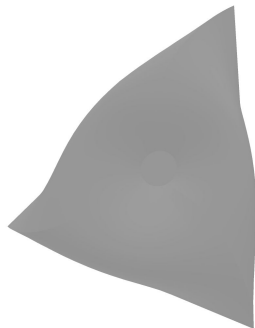
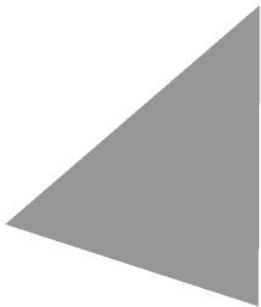




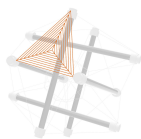
# Accessories Design



## AIDtensegrity - Accessories Design



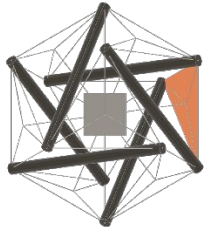
# AIDtensegrity - Accessories Design



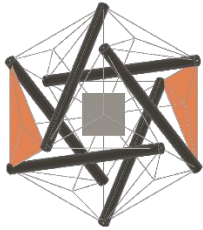




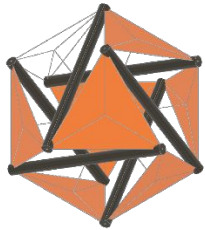
## AIDtensegrity - Accessories Design



**Design 1 (Rotate)**



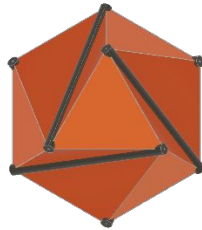
**Design 2 (Not Rotate)**



**Design 3 (Rotate)**



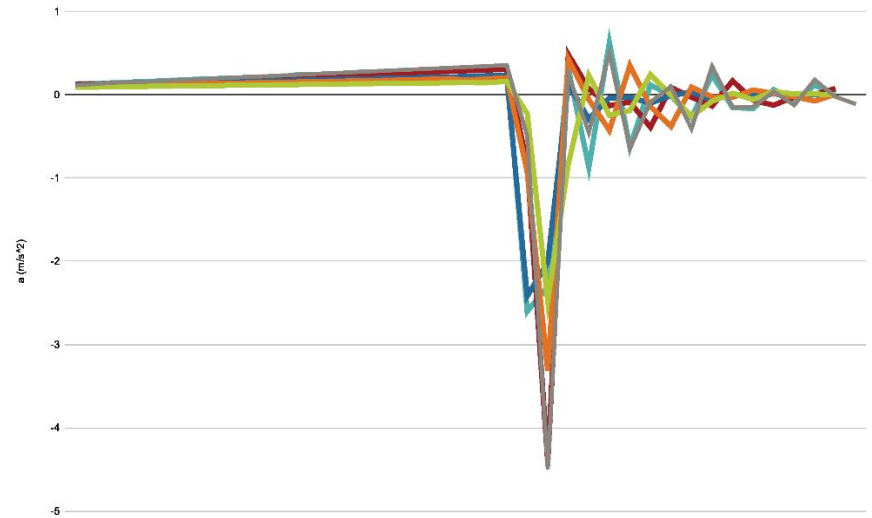
**Design 4 (Rotate)**



**Design 5 (Rotate)**

- Design 1 (One Surface)
- Design 2 (Two Surface)
- Design 3 (Miss One Surface)
- Design 4 (Half Cover)
- Design 5 (Full Cover)
- Web-Based Design

Force Analysis of Various Designs with Constant Mass and Size





# Our Product







# Outlook Statement



# AIDtensegrity - Contribution and Vision

## Contribution

1

In this project, we explore different kinds of tensegrity design structures and test them to evaluate whether they can be used in aerial delivery.

2

We developed a logistics simulation environment in Grasshopper for users to test the model, simulate the dropping action, and collect data.

3

We refined the fabrication process to enable users to build the model more easily and efficiently.

## Vision

1

We aim to explore various design attachments to assess how they enhance performance across different environments.

2

We aim to find a better, automated fabrication process to eliminate the need for manual labor and reduce user difficulty.

3

Physically testing the model by dropping it from a height of 30 meters.



Thank You !